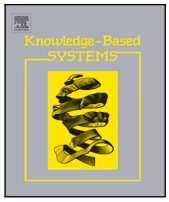
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# Enabling controllable table-to-text generation via prompting large language models with guided planning

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ABSTRACT

Recently, Large Language Models (LLMs) has demonstrated unparalleled capabilities in understanding and generation, hence holding promising prospects for applying LLMs to table-to-text generation. However, the generation process with LLMs lacks a high degree of controllability, which hinders the utilization of LLMs for table-to-text generation. In this paper, we introduce Poised, an effective method that prompts LLMs with guided planning to achieve controllable table-to-text generation. Specifically, we first employ prefix-tuning on BART to derive a plan from the given table. Then, we combine the plan with guided instructions to create a comprehensive prompt, which is later input into LLMs to generate the description of the table. Experiments across three domains of the few-shot *Wiki* dataset show that Poised achieves or approaches a plan completion rate of 100%, with an average hallucination frequency of less than 10%. Furthermore, Poised allows for fine-grained control over the generated content by intentionally modifying the prompt, enabling precise control over aspects such as attribute realization order.

## 1. Introduction

As a subfield of Natural Language Generation (NLG), table-to-text generation focuses on converting structured tables into corresponding textual descriptions [1]. It finds application in diverse real-world scenarios that involve handling substantial amounts of structured data, including sports news reports [2], weather forecasting [3], and health reports [4].

Recently, the new generation of LLMs, exemplified by models like InstructGPT [5] and Vicuna [6], has showcased a remarkable ability to align with user intent. This notable strength positions these models as promising tools to address a variety of practical applications. Consequently, there is great potential to leverage LLMs for table-to-text generation.

However, directly applying LLMs for generation exhibits limited controllability, thus falling short of the high controllability demands of table-to-text generation systems. In this context, controllability refers to the ability to manage the content of the generated text. For instance, when the table in Fig. 1 is provided as input, practical systems should filter out trivial attributes like *length* and *label*, and use important ones (highlighted in color) like *name* and *songwriters* to generate content according to human preferences. Additionally, it is preferable for such systems to achieve fine-grained control, such as controlling the attribute realization order. Therefore, the exploration of methods to enable controllable generation with LLMs carries significant value.

In fact, the limited controllability of LLMs originates from the inherent complexity of the task, which is somewhat beyond the current capabilities of the LLMs. To be specific, the table-to-text generation task requires models to deduce key attributes from the table and then use them in description generation. However, as shown in the description generated by ChatGPT in Fig. 1, LLMs struggle to infer significant attributes effectively without fine-tuning, thus resulting in uncontrollable generation.

To address this issue, we try to transfer the challenging task to what LLMs can handle perfectly. More precisely, we decompose the end-to-end generation into two steps: firstly, generating explicit plans for LLMs, and then guiding LLMs with plans during generation. This idea offers two key advantages. Unlike the chain-of-thought approach [7] that emphasizes better eliciting the reasoning capabilities of LLMs, aligning LLMs with the plan during generation alleviates the demand for sophisticated reasoning skills of LLMs, thus reducing the likelihood of hallucinations and enhancing the controllability during generation. Furthermore, the two-step design allows users to customize details (such as plans) for precise control over the generated content, thereby augmenting the flexibility during generation.

Motivated by the aforementioned idea, we present **Poised** for controllable table-to-text generation. Specifically, we first utilize prefix-tuning to train BART as a content planner, which enables us to obtain

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0950-7051/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

| Attribute | Value |
|---|---|
| Name | Perfect |
| Single_by | Ed Sheeran |
| From_album | ÷ |
| Released | 2017-09-26 |
| Format | CD single · digital download |
| Length | 4:23 |
| Songwriters | Ed Sheeran |
| Genre | Pop |
| Producers | Ed Sheeran · Will Hicks |
| Label | Asylum, Atlantic |

**Reference:** "Perfect" is a song by English singer-songwriter Ed Sheeran from his third studio album, ÷ (2017).

**ChatGPT:** The table provides key details about the song "Perfect" by Ed Sheeran, including its release date on September 26, 2017, duration of 4 minutes and 23 seconds, ···, as well as its labels (Asylum and Atlantic).

**Fig. 1.** A data sample from *Wiki Songs* domain. Directly using LLMs like ChatGPT for generation exhibits limited controllability, making it difficult to produce descriptions aligned with the reference.

high-quality plans even with limited training data and computing resources. Subsequently, we extract key attributes from the input table according to its plan, creating a derived table. We then combine this derived table with one or two examples from the training data, along with guided instructions, to compose a comprehensive prompt. Finally, we input the prompt into LLMs to generate the description of the given table.

We evaluate Poised on three domains (*Humans*, *Books* and *Songs*) of the few-shot *Wiki* dataset [8]. In terms of automatic metrics, Poised demonstrates comparable, and in some cases superior, performance to previous state-of-the-art methods. Moreover, extensive controllability evaluation highlights that Poised is capable of generating texts that faithfully align with the provided plans, affirming the efficacy of our proposed approach for controllable generation. Furthermore, our experiments reveal that intentional modifications to the prompt empower Poised to achieve fine-grained control over the generated content such as attribute realization order.

In summary, our contributions are as follows:

- We offer a new perspective on adapting LLMs to challenging tasks. In contrast to the chain-of-thought approach which elicits the reasoning capabilities of LLMs, transforming complex tasks into ones that LLMs can handle well is a simple yet effective solution.
- Building on our perspective, we present Poised for controllable table-to-text generation that performs well in few-shot settings.
- Experiments on the few-shot *Wiki* dataset show that Poised demonstrates a significant improvement in controllable generation. It achieves or comes close to a plan completion rate of 100%, with an average hallucination frequency of less than 10%.

## 2. Related work

*Few-shot table-to-text generation.* Table-to-text generation has been a long-standing NLG task that has been extensively researched. Recently, few-shot table-to-text generation has gained more focus for its relevance to real-world scenarios with limited annotated data. Ma et al. [9] initially explore low-resource table-to-text generation with a pipeline approach called Pivot, which comprises two steps: key fact prediction and surface realization. After Pivot, subsequent works employ Pre-trained Language Models (PLMs) for improved performance. Chen et al. [8] introduce copy mechanism to GPT-2 [10] for accurate content selection. Gong et al. [11] exploit multi-task learning with two auxiliary tasks to improve the structure-encoding ability of GPT-2. Su et al. [12], Guo et al. [13] and Feng et al. [14] use an information retrieval (IR) system to infuse knowledge from large unlabeled corpus to PLMs. However, the performance of such methods relies heavily on highly correlated large-scale corpora, which limits their practical applicability. Zhao et al. [15] modify the Transformer architecture to achieve a two-level attention mechanism and implicit planning. Luo et al. [16] prepend a task-specific prefix to adapt to downstream tasks and an input-specific prefix to perform generation planning. Wang et al. [17] utilize Cycle Training to enhance the faithfulness of generated descriptions for out-of-domain data. Li et al. [18] provide a unified representation capable of handling various forms of structured data. Lately, Zhao et al. [19] and Lorandi et al. [20] have investigated the performance of LLMs in few-shot table-to-text generation tasks.

In contrast to previous works that consider planning [9,15,16], our method steers LLMs to align with the plans during generation, enabling the better exploitation of planning and consequently yielding a distinct improvement in controllability. Besides, our work provides a new perspective for LLMs to adapt to complex tasks.

*Prompting for generation.* According to [21], there is a significant transformation in the learning of NLP models, wherein the "pre-train and fine-tune" paradigm is gradually being replaced by the "pre-train, prompt, and predict" paradigm.

One line of work in this field, which is referred to as *continuous prompts* (a.k.a *soft prompts*), prepends a sequence of continuous vectors as prompts. Lester et al. [22] propose Prompt Tuning, which inserts trainable vectors before input embeddings to achieve efficient adaptation to downstream tasks. Li and Liang [23] propose prefix-tuning, which injects vectors into the multi-head attention module of the Transformer, resulting in improved fitting capability. Another line of work uses natural language as prompts, which is called *discrete prompts* (a.k.a *hard prompts*). GPT-3 [24] introduces in-context learning, which employs human-designed prompts for generation tasks within few-shot settings. Recently, researchers have fine-tuned LLMs with human feedback to align them with user intent. As a result, users can utilize instructive prompts to address practical problems with LLMs such as InstructGPT [5] and Vicuna [6]. More recent studies have employed chain-of-thought [7] or retrieval-augmented [25] techniques to further enhance the generation performance of LLMs.

To the best of our knowledge, our work is the first to leverage the new generation of LLMs for controllable few-shot table-to-text generation. By transforming the whole table-to-text generation task into the one suited for LLMs, we have enabled controllable table-to-text generation.

## 3. Methodology

Fig. 2 depicts the overview of Poised. We start by presenting the problem definition in Section 3.1. Next, we elaborate on the plan extraction stage in Section 3.2 and the prompt-guided generation stage in Section 3.3.

### 3.1. Problem definition

In the plan extraction stage, we input a Table $T$, which consists of $n$ attribute–value pairs$\{a_i : v_i\}_{i=1}^{n}$, to obtain a Plan $P$. Subsequently, in the prompt-guided generation stage, we assemble a Comprehensive Prompt $C$ based on $P$ and then input $C$ into the LLMs to generate the output $Y$.

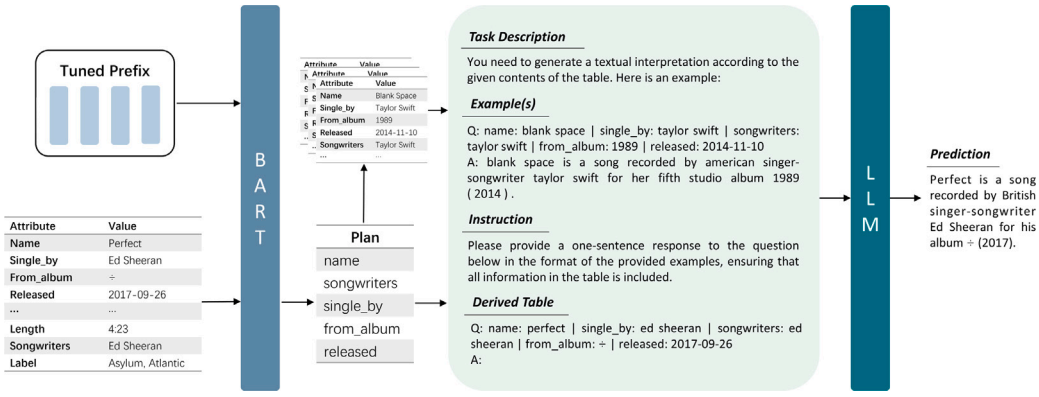



**Fig. 2.** The overview of Poised. Poised employs two consecutive stages for table-to-text generation, namely plan extraction and prompt-guided generation. In the plan extraction stage, Poised uses a tuned prefix to steer BART in deriving a generation plan from the given table. In the prompt-guided generation stage, it combines the plan with guided instructions to create a comprehensive prompt, which is later input into LLMs to generate the description of the table.

### 3.2. Plan extraction

In this stage, we adopt BART-large [26] as our base model. Then, we employ prefix-tuning [23] to train a tuned prefix that guides BART in deriving a generation plan from the given table. As an example, taking the table in Fig. 2 as input, this stage generates the planning sequence "*name songwriters single_by from_album released*" to guide the subsequent generation process.

Prefix-tuning is a parameter-efficient transfer learning method that adapts to downstream tasks by utilizing a set of trainable vectors while keeping the model parameters frozen. Specifically, prefix-tuning inserts a sequence of trainable vectors (assumed to have a length of $l$) at the beginning of the $\boldsymbol{K}$ matrix and $\boldsymbol{V}$ matrix respectively in the multi-head attention module of each Transformer block. Formally, we represent prefixes as $\boldsymbol{P}_k$ and $\boldsymbol{P}_v$. First, we concatenate $\boldsymbol{P}_k$ with $\boldsymbol{K}$ and $\boldsymbol{P}_v$ with $\boldsymbol{V}$. Then, we perform multi-head attention on the modified keys and values. In this way, the calculation of $head^i$ becomes:

$$\begin{aligned} head^i &= Attention(\boldsymbol{Q}^i, \boldsymbol{K}^i, \boldsymbol{V}^i) \\ &= softmax(\frac{\boldsymbol{Q}^i[\boldsymbol{P}_k^i : \boldsymbol{K}^i]^T}{\sqrt{d_k}})[\boldsymbol{P}_v^i : \boldsymbol{V}^i], \end{aligned} \tag{1}$$

where $d_k$ denotes the hidden dimension of each head, and $\boldsymbol{P}_k^i$, $\boldsymbol{P}_v^i \in \mathbb{R}^{l \times d_k}$ denote a part of $\boldsymbol{P}_k$, $\boldsymbol{P}_v$ respectively. Here, we skip the introduction of the multi-head attention mechanism. Please refer to [27] for detailed information.

The reasons for employing prefix-tuning to extract plans are as follows: Firstly, prefix-tuning delivers impressive performance in scenarios with limited training data. This advantage is evident when compared to rule-based extraction methods, which often struggle to strike a balance between plan quality and the cost of expert-designed rules. Secondly, with prefix-tuning, we only need to store a single copy of PLMs with fewer parameters for multiple downstream tasks. This contrasts with methods like employing LLMs for planning, highlighting prefix tuning's efficiency. Moreover, prefix-tuning exhibits greater flexibility compared to other parameter-efficient transfer learning methods, such as Adapter [28] and LoRA [29]. This is because prefix-tuning enables batching multiple users' queries even when they are supported by different prefixes under the personalization setting. In conclusion, in contrast to other methods, prefix-tuned BART offers better efficiency and flexibility.

In this work, we add prefixes to both the encoder and decoder of BART. The number of parameters associated with the prefixes is less than 1% of the total parameters in BART. We tune the prefixes with table-plan pairs from the few-shot trainset during training. Given a training instance $(\boldsymbol{T}, \boldsymbol{P})$, we use cross-entropy as the learning objective:

$$\mathcal{L}_{LM} = -\sum_{i=1}^{|\boldsymbol{P}|} \log p_\phi(\boldsymbol{P}_i | \boldsymbol{P}_{<i}, \boldsymbol{T}), \tag{2}$$

where $\phi$ denotes the parameters introduced by prefix-tuning. Note that the BART parameters $\theta$ are frozen during the training steps.

Additionally, to ensure the stability of prefix-tuning in few-shot settings, we first initialize the prefix with activations of task-specific real words, such as "table-to-text generation". Secondly, we adopt the reparameterization strategy to prefixes. Li and Liang [23] pointed out that reparameterizing prefixes with an MLP makes the training less sensitive to the choice of hyperparameters. Inspired by this, we use two different MLPs to reparameterize the prefixes of the encoder and decoder respectively.

### 3.3. Prompt-guided generation

The prompt-guided generation stage aims to utilize LLMs for generating controlled content. In this stage, we start by creating a comprehensive prompt based on the generated plan from the previous stage. Next, we feed this prompt into LLMs to obtain the description of the given table.

The design of the prompt is an important factor to achieve controllability. Firstly, we leverage the alignment capability of LLMs with user intent, enabling them to generate content in accordance with the provided plans. Furthermore, by incorporating specific instructions, we mitigate the potential for hallucination in LLMs, which further enhances the control over the generation.

Here, we provide a detailed explanation of the comprehensive prompt. As shown in Fig. 2, the prompt comprises four parts, namely *Task Description*, *Example(s)*, *Instruction*, and *Derived Table*.

**Task Description.** Brown et al. [24] suggest the necessity of a natural language task description within in-context learning. Experiments in their work illustrate that the addition of a task description leads to improved performance for models. Therefore, we begin the comprehensive prompt with a task description that clarifies the objective of table-to-text generation.

**Example(s).** Task examples provide crucial reference samples for LLMs. In our work, we identify task examples by comparing the plans of the input tables with those in the few-shot trainset, which contains only dozens of data pairs. We employ an exact matching strategy. If the plan of the input table is found in the trainset, we use the corresponding data pairs as task examples. If not, we select the most informative data pairs as task examples.

**Instruction.** In order to leverage the alignment capability of LLMs with user intent, We add task-specific instructions to the prompt, aiming to achieve fine-grained control over the generated content. Specifically, there are three types of instructions: (1) *Length Instruction*, which limits the generated content to a single sentence; (2) *Format Instruction*, which guides LLMs to refer to the format of the table realization in examples; (3) *Coverage Instruction*, which directs LLMs to cover all the attributes in the plan during generation.

**Derived Table.** The derived table is obtained by extracting attributes from the original input table based on its plan. There are two purposes for simplifying the table: firstly, to allow LLMs to generate content in a controlled manner by only attending to important attributes, and secondly, to accelerate the inference time.

## 4. Experiments

In this section, we first introduce the datasets and experiment setup in Section 4.1. Then we present the results of the automatic evaluation in Section 4.2. Due to the limitations of automatic metrics in evaluating model controllability, we use human evaluation to assess the controllability of Poised in Section 4.3. Additionally, we analyze several characteristics of Poised through qualitative study in Section 4.4.

### 4.1. Datasets and experiment setup

*Datasets.* We evaluate Poised on three domains (*Humans*, *Books* and *Songs*) of the *Wiki* dataset [8]. Due to the high cost of utilizing LLMs to produce results in all few-shot settings provided by the dataset, we primarily concentrate on conducting experiments in a few-shot setting with a training set size of 100. Consistent with previous works, the test set sizes are 13 587, 5252, and 11 879 for *Humans*, *Books*, and *Songs*, respectively. Furthermore, we acquire the supervised data for the plan extraction stage through manual annotation, with 100 instances per dataset. Note that we do not use the data from the validation set in both stages to better simulate real-world few-shot scenarios.

*Experiment details.* In the plan extraction stage, the base model is BART-large, which has 12 Transformer layers and 16 heads for both the encoder and decoder. We set the prefix length to 10. During training, the maximum training steps are set to 300, and the batch size is set to 10. We utilize the Adam optimizer with a learning rate of 5e−5. Since we do not use the validation set, we allow the model to overfit on the training set. In the prompt-guided generation stage, we choose LLaMa-13B [30], Vicuna-7B, Vicuna-13B, and GPT-3 (text-davinci-003) as our generation model. The experiments involving GPT-3 are conducted using the OpenAI API,[1] while the remaining experiments are performed on 1 to 3 RTX 3090Ti GPUs.

*Baselines.* We compared our method with strong baseline models that do not inject external knowledge. There are two categories of baselines. The first category is the previous state-of-the-art few-shot table-to-text generation methods, including **Pivot** [9], **Switch-GPT** [8], **TableGPT** [11], and **PCG** [16]. The second category is BART-large tuned by fine-tuning and prefix-tuning, denoted as **BART-large (FT)** and **BART-large (PT)** respectively.

### 4.2. Automatic evaluation

First, we use BLEU-4 [31] to automatically compare Poised (with Vicuna-13B as the generator) to the strong baselines. The results of the three domains are shown in Table 1.

According to Table 1, Poised achieves the best performance in the *Books* and *Songs* domains, while it performs on par with the baselines in the *Humans* domain, except for PCG. Such results suggest that Poised is capable of producing fluent descriptions as the strong baselines do. Through data analysis, we have observed that the tables in *Humans* domain contain less detailed information compared to the other two domains, in which case models are occasionally required to infer attributes such as occupation and nationality of a person. Unfortunately, Poised filters out the redundant attributes and does not fine-tune LLMs in the prompt-guided generation stage, which leads to its suboptimal performance in *Humans* domain. For further analysis and the solution to this issue, please refer to Section 4.3.

**Table 1**
BLEU-4 results on the complete test set of *Humans*, *Books* and *Songs* with a training set size of 100.

| Methods | Domains | | |
|---|---|---|---|
| | Humans | Books | Songs |
| Pivot[a] | 18.7 | 24.9 | 28.0 |
| Switch-GPT[a] | 29.5 | 36.2 | 37.2 |
| TableGPT[a] | 34.5 | 37.3 | 37.8 |
| PCG[a] | **43.3** | 36.9 | 41.7 |
| BART-large (PT) | 36.8 | 35.9 | 38.0 |
| BART-large (FT) | 38.5 | 38.0 | 40.1 |
| Poised (Vicuna-13B) | 34.7 | **39.2** | **41.8** |

[a] Indicates that the results of the model are copied from the original paper.

**Table 2**
BLEU-4 results of different LLMs on the trimmed test set, which consists of the first 1000 instances from the complete dataset.

| LLMs | Domains | | |
|---|---|---|---|
| | Humans | Books | Songs |
| LLaMa-13B | 32.3 | 40.0 | **45.7** |
| Vicuna-7B | 34.0 | 40.6 | 45.3 |
| Vicuna-13B | 35.1 | **40.8** | 41.6 |
| GPT-3 (davinci) | **36.2** | 40.6 | 44.6 |

Furthermore, we conducted experiments using different LLMs as the generator for the second stage of Poised. Due to the high computational cost associated with LLM-based text generation, we trim the test set by keeping only the first 1000 instances from each of the three domains for evaluation. The results are presented in Table 2.

In Table 2, the performance of Vicuna-13B on the first 1000 instances closely aligns with its performance on the complete test set in Table 1, suggesting that the trimmed test set shares a similar distribution with the complete test set. Additionally, we notice that LLMs achieving the best performance differ across three domains. Moreover, LLMs with various parameters exhibit comparable performance across three domains. This implies that employing a smaller LLM for table-to-text generation is feasible, emphasizing the practicality of our approach.

### 4.3. Intrinsic evaluation

We conduct extensive experiments with human evaluation[2] to assess the controllability of Poised. Specifically, all the results in this section are obtained by evaluating predictions of the first 100 tables from the test set in their respective settings.

#### 4.3.1. Overall controllability study

We start by introducing the evaluation metrics: (1) Number of missed attributes (**#miss**): This metric provides a macroscopic perspective by counting the total number of attributes from plans that are not realized among the 100 predictions. (2) Completion rate (**c. rate**): This metric offers a microscopic perspective by calculating the attribute realization rate for each individual result and then taking the average across the 100 predictions. (3) Number of contradictions (**#con**): This metric measures the total number of facts that are contradictory to the table or reference out of the 100 predictions. For instances of unrealized attributes in plans and contradictions, please refer to Appendix C.

[1] https://platform.openai.com/playground.

[2] We recruited 15 postgraduate volunteers proficient in English skills for human evaluation. Before evaluation, we inform human judges of the evaluation criteria through documents and cases. During evaluation, each output is presented to three human judges and is provided along with its corresponding complete table and prompt.

**Table 3**
Results of the overall controllability study on three domains.

| LLMs | Humans | | | Books | | | Songs | | |
|---|---|---|---|---|---|---|---|---|---|
| | #miss ↓ | c. rate | #con ↓ | #miss ↓ | c. rate | #con ↓ | #miss ↓ | c. rate | #con ↓ |
| LLaMa-13B | 8 | 98.02% | 14 | 2 | 99.17% | 10 | 5 | 98.81% | 8 |
| Vicuna-7B | 4 | 99.05% | 13 | 1 | 99.75% | **6** | **0** | **100%** | 6 |
| Vicuna-13B | 4 | 98.80% | **11** | **0** | **100%** | 8 | 1 | 99.75% | **5** |
| GPT-3 (davinci) | **0** | **100%** | 12 | **0** | **100%** | **6** | **0** | **100%** | **5** |

**Table 4**
Results of text fidelity comparison between Poised and previous state-of-the-art methods in *Humans* domain.

| Method | #avg. con ↓ |
|---|---|
| TableGPT[a] [11] | 0.60 |
| P2G[a] [12] | 0.47 |
| AMG[a] [15] | 1.75 |
| Poised (Vicuna-13B) | **0.11** |

[a] Indicates that the results of the model are copied from the original paper.

The results of different LLMs are presented in Table 3. Firstly, we observe that LLaMa-13B performs inferior to other LLMs in terms of the three controllability metrics, with more errors in the generated text. We suggest that the disparity arises from the training strategies employed by Vicuna and GPT-3, which involve aligning with user intent. These strategies enhance the understanding capabilities of LLMs, leading to improved controllability over the generated content. Secondly, GPT-3 outperforms all other LLMs. It exhibits the fewest contradictions with the ground truth, and achieves a 100% attribute completion rate across all three datasets, showcasing its exceptional controllability. Such results indicate that LLMs with a larger number of parameters tend to possess stronger comprehension and reasoning abilities.

Furthermore, we compare the text fidelity of Poised against methods that share similar experimental settings with our work. According to Table 4, Poised makes an average of one factual error for every ten predictions in *Humans* domain. The comparison reveals that Poised (Vicuna-13B) surpasses previous state-of-the-art models by a significant margin in terms of text fidelity. We attribute this performance to the exceptional controllability of our method. The fine-grained control during generation allows Poised to align the generated content with the plan, significantly reducing the occurrence of hallucinations.

Admittedly, the experiments reveal that Poised is not entirely controllable during the generation process. It may omit certain attributes mentioned in the plan or introduce contradictions against the ground truth. This observation is particularly noticeable in *Humans* domain, which is in line with the performance gap between Poised and top models in terms of BLEU-4 in this domain. In the process of human evaluation, we find that the reasons for the errors in Poised's generation mainly come from two aspects. Firstly, there are instances where the plan and examples do not match, resulting in the LLMs' failure to find relevant instances for reference during generation. Secondly, some attributes in the table lack key information, which causes LLMs to reason about the missing content when generating, thus increasing the likelihood of hallucinations. For instance, tables in *Humans* domain often lack attributes such as "occupation" and "nationality", while tables in *Books* and *Songs* domains frequently miss the "publication_date" attribute (or "released_date" in *Songs*), and the "series" attribute often lacks the specific serial number.

Here, we propose an effective solution, which is providing LLMs with more informative tables. In the case study of *Humans* domain shown in Fig. 3, the original table lacks the "occupation" attribute, which leads LLMs to hallucinate the wrong occupation during prediction. However, when an informative table is provided to LLMs, Poised generates the correct content in a controlled manner based on the new plan. In practical applications, ensuring the informativeness of tables can be achieved through predefined rules, thus making it a feasible solution.

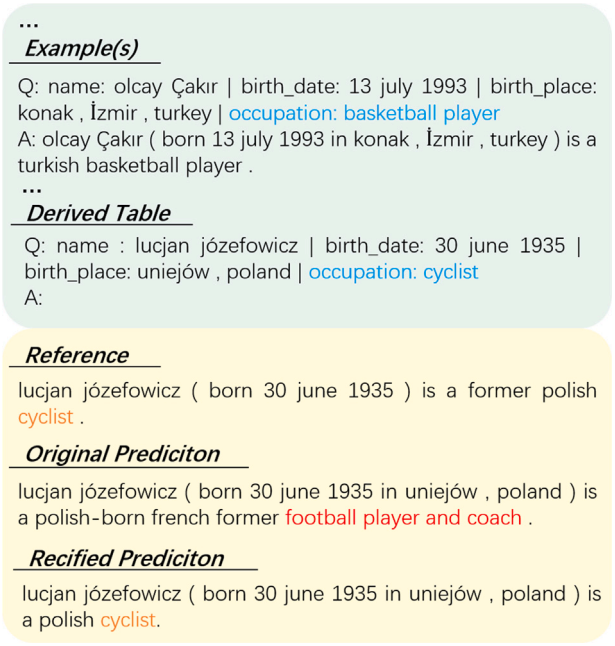



**Fig. 3.** Case study in *Humans* domain. Text in blue denotes the additional information, in red indicates incorrect facts, and in orange represents the corresponding correct facts.

In the following part of this section, we conduct ablation studies to examine the effects of Instruction, Example(s), and Derived Table (corresponding to the planning phase in the first stage) respectively.

#### 4.3.2. One-step vs. Two-step

To validate the effectiveness of task transferring, we compare Poised (which decomposes the generation into a two-step process) with the one-step approach (which directly utilizes LLMs for generation).

In the one-step scenario, tables in prompts are not explicitly derived, and the generation model needs to learn implicit planning from the examples to generate descriptions corresponding to the complete tables. To ensure experimental consistency, both settings utilize Vicuna-13B as the generation model, with two examples provided, and the examples' tables are the complete versions in the corresponding one-step setting. What is more, we modified the Coverage Instruction to "ensuring that the attribute selection is consistent with references in examples" to guide LLMs to derive implicit plans from examples in the one-step setting. Since the planning in the one-step scenario is implicit, it is challenging to evaluate its impact on the generated content. Therefore, during the evaluation, we excluded plan-related metrics to focus solely on other factors.

The results in Table 5a indicate that Poised outperforms the one-step variant overall. Firstly, the one-step variant produces more factual errors, suggesting that LLMs struggle to effectively learn implicit planning, thus leading to more errors in the generated output. Furthermore, the one-step variant has a longer average generation time. This is partly due to the increased prompt length caused by using the complete table,

**Table 5**
Ablation studies of Poised in *Books* domain.

(a) A comparison between one-step generation and two-step generation. The "**avg. time**" metric represents the average time taken by the model to make 100 predictions on the test set.

| Method | #con ↓ | avg. time |
|---|---|---|
| one-step | 11 | 18.4 s |
| two-step (ours) | 8 | 3.8 s |

(b) An ablation study on three types of instructions.

| variants | #miss ↓ | c. rate | #con ↓ |
|---|---|---|---|
| Poised (Vicuna-13B) | **0** | **100%** | **8** |
| *-format instruction* | 4 | 98.92% | 14 |
| *-length instruction* | 2 | 98.50% | 10 |
| *-cover instruction* | 4 | 99.10% | 13 |

(c) A study on the number of examples.

| #example | #miss ↓ | c. rate | #con ↓ | avg. time |
|---|---|---|---|---|
| 0 | 6 | 98% | 6 | 7.4 s |
| 1 | 5 | 98.55% | 8 | 6.6 s |
| 2 | 0 | 100% | 8 | 3.8 s |
| 3 | 0 | 100% | 8 | 4.5 s |
| 4 | 0 | 100% | 9 | 5.2 s |
| 5 | 0 | 100% | 7 | 5.6 s |

resulting in additional encoding time. Moreover, the one-step variant's inferior understanding of instructions causes longer predictions, subsequently leading to increased decoding time. In conclusion, the experiment demonstrates the necessity of an additional planning step. By performing planning before generating the text, we achieve a more controllable generation process.

#### 4.3.3. Ablation study of instruction

We conduct an ablation study on three types of instructions in *Books* domain, and the results are illustrated in Table 5b.

According to Table 5b, the performance of the model across three controllability metrics decreases when any of the instructions are removed. Specifically, the format and cover instructions have the greatest impact on the model. In the absence of these two instruction types, the predictions exhibit more factual errors and failures in attribute realization. On the other hand, the removal of the length instruction has a relatively minor effect on the model. However, LLMs tend to produce longer text without the length instruction, thereby increasing the likelihood of hallucinations. It is noteworthy that the **c.rate** metric plays a significant role in the ablation study. It applies a substantial penalty when multiple unrealized attributes are present in the same plan. This is evident in the "*-length instruction*" variant, which produces fewer failures in attribute realization, but has a lower completion rate compared to other variants.

#### 4.3.4. Numbers of examples

We investigated the influence of the number of examples on the performance of Poised. Here we use Vicuna-13B as the generator, and the results are presented in Table 5c.

In terms of three controllable metrics, when the number of examples is two or more, Poised demonstrates stable controllability, achieving a 100% completion rate with a low number of factual errors. Regarding the average time taken during generation, we observe a decrease followed by an increase as the number of examples increases. The minimum average generation time of 3.8 s is observed when two examples are provided. This is because with fewer examples, Poised has limited control over the generated content, including its length. A higher frequency of producing overly long prediction results in a longer decoding time. Conversely, as the number of examples increases, the length of the input prompt increases, resulting in additional encoding time for LLMs. Overall, the experiment shows that Poised exhibits strong control capabilities and high generation efficiency when two examples are given.

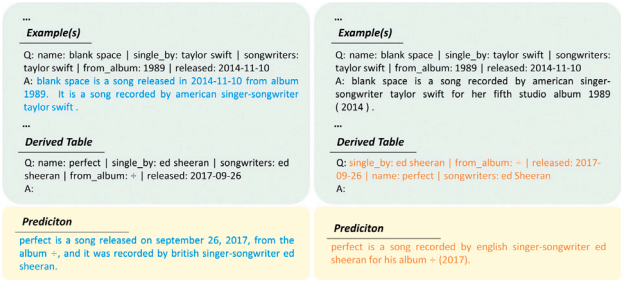



**Fig. 4.** Case studies on attribute realization order.

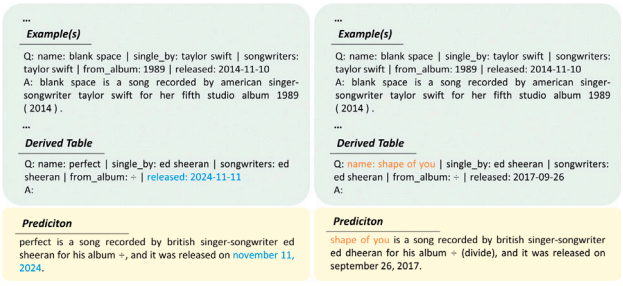



**Fig. 5.** Case studies on structure awareness ability of LLMs.

### 4.4. Discussion

In this section, we qualitatively explore some characteristics of Poised through case studies.

**What determines the attribute realization order?** In this part, we explore the factors that determine the attribute realization order in the prompt of Poised. We propose two hypotheses: firstly, the attribute order in the reference of Example(s) determines the attribute order in the predictions; secondly, the attribute order in the Derived Table influences the attribute order in the predictions. We validate these hypotheses through case studies. Specifically, we manipulate the attribute order in the reference of Example(s) or the attribute order in the Derived Table and compare the resulting outputs.

The results of the case studies on attribute realization order are depicted in Fig. 4. In the left figure, the attribute order is altered in the reference of Example(s), while in the right figure, the attribute order is shuffled in the Derived Table. The results demonstrate that Poised is not sensitive to the attribute order in the Derived Table Instead, it relies on the attribute order specified in the reference of Example(s) during the generation process. This implies that Poised can achieve fine-grained control over the generated content by intervening in the attribute order in the reference of the Examples.

**Can LLMs understand the structure of tables?** Unlike natural language, tabular data carries abundant structural information. One of the challenges in the field of table-to-text generation is enabling models to effectively learn the structure of tables. In this study, we conduct experiments to explore whether LLMs can grasp the structure of tables. Specifically, we randomly replace attribute values in the Derived Table and observe how LLMs handle these substitutions.

The results of the case studies on the structure awareness ability of LLMs are shown in Fig. 5. In the left figure, the value of the "released" attribute is modified, while in the right figure, the value of the "name" attribute is altered. The results indicate that when attribute values are replaced, LLMs demonstrate the capability to accurately substitute the old attribute values with the new ones during the generation process. This demonstrates that the controllable generation of Poised is built upon an understanding of the underlying structure of the tables.

**Table A.6**
ROUGE-L results on the test set of the Wiki dataset.

| Methods | Domains | | |
|---|---|---|---|
| | Humans | Books | Songs |
| Pivot[a] | 6.9 | 13.3 | 16.2 |
| Switch-GPT[a] | 16.2 | 23.1 | 28.6 |
| TableGPT[a] | 20.6 | 25.4 | 29.4 |
| PCG[a] | – | – | – |
| BART-large (PT) | 22.4 | 27.1 | 28.1 |
| BART-large (FT) | **24.0** | 27.5 | 29.6 |
| Poised (Vicuna-13B) | 19.8 | **28.9** | **29.8** |

[a] Indicates that the results of the model are copied from the original paper.

**Table B.7**
Case studies on model scalability of Poised.

**Humans Domain**
You need to convert the table into a corresponding text description, and the description needs to contain all the contents of the table. Here is an example:
Q: name: olcay Çakır | birth date: 13 july 1993 | birth place: konak, İzmir, turkey | occupation: basketball player
A: olcay Çakır (born 13 july 1993 in konak, İzmir, turkey) is a turkish basketball player.
Please provide a one-sentence response to the question below in the format of the provided examples, ensuring that all information in the table is included.
Q: name : lucjan józefowicz | birth date: 30 june 1935 | birth place: uniejów, poland | occupation: cyclist
A:

**Mixtral-8 × 7B-Instruct:**
Lucjan Józefowicz (born June 30, 1935, in Uniejów, Poland) is a Polish cyclist.
**CPT-4 Turbo:**
lucjan józefowicz (born 30 june 1935 in uniejów, poland) is a polish cyclist.

**Books Domain**
You need to convert the table into a corresponding text description, and the description needs to contain all the contents of the table. Here is an example:
Q: name: once an eagle | genre: war | country: united states | author: anton myrer | publication date: 1968
A: once an eagle is a war novel by american author anton myrer published in 1968.
Please provide a one-sentence response to the question below in the format of the provided examples, ensuring that all information in the table is included.
Q: name: all passion spent | genre: fiction | country: united kingdom | author: vita sackville-west | publication date: 1931
A:

**Mixtral-8 × 7B-Instruct:**
All Passion Spent is a work of fiction by British author Vita Sackville-West, published in 1931.
**CPT-4 Turbo:**
all passion spent is a fiction novel by british author vita sackville-west published in 1931.

**Songs Domain**
You need to convert the table into a corresponding text description, and the description needs to contain all the contents of the table. Here is an example:
Q: name: dear doctor | genre: rock and roll | single by: the rolling stones | released: 6 december 1968 | from album: beggars banquet
A: dear doctor is a song by english rock and roll band the rolling stones featured on their 1968 album beggars banquet.
Please provide a one-sentence response to the question below in the format of the provided examples, ensuring that all information in the table is included.
Q: name: declaration | genre: gospel | single by: kirk franklin | released: october 23, 2007 | from album: the fight of my life
A:

**Mixtral-8 × 7B-Instruct:**
Declaration is a gospel single by Kirk Franklin, released on October 23, 2007, as part of his album The Fight of My Life.
**CPT-4 Turbo:**
declaration is a song by american gospel artist kirk franklin featured on his 2007 album the fight of my life.

**Table B.8**
Case studies on domain scalability of Poised.

**E2E Dataset (Restaurant Domain)**
You need to convert the table into a corresponding text description, and the description needs to contain all the contents of the table. Here is an example:
Q: name: Green Man | eatType: restaurant | food: Japanese | area: riverside | familyFriendly: yes | near: Express by Holiday Inn
A: Green Man is a family friendly restaurant that offers Japanese food. It is near Express by Holiday Inn in the area of riverside.
Please provide a one-sentence response to the question below in the format of the provided examples, ensuring that all information in the table is included.
Q: name: The Cricketers | eatType: restaurant | food: Chinese | area: city centre | familyFriendly: no | near: All Bar One
A:

**Vicuna-13B:**
The Cricketers is a restaurant that offers Chinese food. It is located in the city centre and is not family friendly. It is near All Bar One.
**GPT-3 (davinci)**
The Cricketers is a restaurant that offers Chinese food. It is near All Bar One in the city centre area and is not family friendly.

**SciGen Dataset (Science Domain)**
You need to convert the table into a corresponding text description, and the description needs to contain all the contents of the table. Here is an example:
Q: Metric: AUC@0.1 | Validation: 0.925 | Test: 0.885
A: As shown in the table, our model scored 0.925 on the validation set and 0.885 on the test set for the AUC@0.1 metric.
Please provide a one-sentence response to the question below in the format of the provided examples, ensuring that all information in the table is included.
Q: Metric: AUC@0.05 | Validation: 0.871 | Test: 0.816
A:

**Vicuna-13B:**
As shown in the table, our model scored 0.871 on the validation set and 0.816 on the test set for the AUC@0.05 metric.
**GPT-3 (davinci):**
As shown in the table, our model scored 0.871 on the validation set and 0.816 on the test set for the AUC@0.05 metric.

**LogicNLG Dataset (Open Domain)**
You need to convert the table into a corresponding text description, and the description needs to contain all the contents of the table. Here is an example:
Q: [song: bara hon älskar mig | place: 1st | points: 80], [song: missarna | place: 3rd | points: 65]
A: The song "bara hon älskar mig", which ranked first, scored 15 points higher than "missarna", which ranked third.
Please provide a one-sentence response to the question below in the format of the provided examples, ensuring that all information in the table is included.
Q: [song: hand i hand | place: 5th | points: 55], [song: world wide web | place: 9th | points: 26]
A:

**Vicuna-13B:**
The song "hand i hand" ranked fifth and scored 29 points more than "world wide web", which ranked ninth.
**GPT-3 (davinci):**
The song "hand i hand", which ranked fifth, scored 29 points higher than "world wide web", which ranked ninth.

**Table C.9**
Further explanation of controllability evaluation criteria.

**Derived Table:**
name: once an eagle | publication date: 1968 | genre: war | country: united states | author: anton myrer
**Reference:**
once an eagle is a 1968 war novel by american author anton myrer.

**Instance of unrealized attributes in plans:**
once an eagle is a war novel by american author anton myrer.
**Instance of contradictions:**
once an eagle is a 2005 war novel by american author anton myrer.

## 5. Conclusion

In this paper, we offer a new perspective on adapting LLMs to complex tasks, which proves to be a straightforward and effective solution compared to other solutions. Based on our perspective, we propose Poised for controllable table-to-text generation. Experiments

on the few-shot public dataset reveal that, in contrast to previous state-of-the-art methods, Poised has achieved a notable advancement in controllable table-to-text generation. Admittedly, our work has certain limitations. Firstly, the text generated by Poised is not completely controllable, which can lead to hallucinations and attribute omissions. Secondly, Poised currently focuses exclusively on single-sentence table-to-text generation tasks. In future work, we aim to address these limitations by striving to develop more controllable and longer-form data-to-text generation approaches.

## CRediT authorship contribution statement

**Shuo Zhao:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Xin Sun:** Writing – review & editing, Supervision, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to polish the writing. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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## Appendix A. ROUGE-L results on the Wiki dataset

To further validate Poised's performance, we compare Poised with strong baselines using the ROUGE-L [32] metric on the Wiki dataset's test set. The experimental results are shown in Table A.6.

As shown in Table A.6, Poised achieves the highest ROUGE-L scores in the Books and Songs domains, where the table information is relatively detailed. However, in the Humans domain, where the table information is less detailed, the performance of Poised is inferior to fine-tuned baselines such as TableGPT and BART-large. This result corroborates Poised's performance in terms of the BLEU-4 metric, further demonstrating its capability to generate fluent and high-quality descriptions.

## Appendix B. Studies on scalability of poised

### B.1. Model scalability

Firstly, we validate Poised's model scalability using more modern LLMs on the Wiki dataset. In this section, we choose the open-source Mixtral-Instruct [33] and the closed-source GPT-4 Turbo to conduct the experiment. The results of the case studies are presented in Table B.7.

From Table B.7, we can observe that after replacing the base LLM, Poised is still able to generate text descriptions for the input tables in a controlled manner. This indicates that Poised can adapt to different LLMs, which demonstrates the good model scalability of our method.

### B.2. Domain scalability

Moreover, we validate Poised's domain scalability using table-to-text generation datasets from different domains. In this section, we conduct experiments on three datasets: E2E [34], SciGen [35], and LogicNLG [36]. The results of the case studies with Vicuna-13B and GPT-3 (Davinci) as generators are shown in Table B.8.

The results in Table B.8 show that with changes in the data domain, Poised can still generate controlled descriptions of the input tables. This demonstrates that Poised has good domain scalability and is capable of converting tables from various domains into corresponding descriptions in a controlled manner in real-world scenarios.

## Appendix C. Further explanation of controllability evaluation criteria

The further explanation of the controllability evaluation criteria is shown in Table C.9. The first example represents an instance of unrealized attributes in plans, as it fails to include the "publication date" attribute from the Derived Table. The second example illustrates an instance of contradictions, where the publication date is incorrectly described as 2005, while the table indicates the publication date is 1968. It is important to note that if there are multiple factual contradictions or missing attributes within a generation, they will be counted multiple times in the tally.

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